



Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

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DEPARTMENT OF CSE

A.Y. 2026 – 2027

B.Tech. III Year Odd Semester

COURSE: CNAD (24CND3101)

PROJECT ABSTRACT SUBMISSION

TITLE OF THE PROJECT	ResilioSpace: A Cloud-Native Self-Adaptive and Self-Healing 2D-to-3D Architectural Visualization Platform	
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Abstract

Traditional architectural design workflows often rely on manual interpretation of 2D floor plans, making 3D visualization, design modification, and error correction time-consuming and difficult to manage. The proposed project, **ResilioSpace**, addresses these limitations by developing a cloud-native architectural visualization platform that converts 2D residential floor plans into interactive 3D models. The platform uses computer vision with OpenCV to detect walls, rooms, doors, windows, and other structural elements from uploaded floor-plan images, followed by confidence evaluation to assess the accuracy of reconstruction. Based on the confidence level and detected issues, the system can adapt its processing strategy or provide a manual correction mechanism. The generated 3D models are visualized using Three.js, allowing users to customize architectural elements such as walls, floors, materials, doors, and windows. The platform also supports Vastu rule analysis, version comparison, and design evaluation. The system follows a modular cloud-native architecture using React.js, Three.js, Tailwind CSS, Python, FastAPI, PostgreSQL, Docker, and GitHub Actions, with Prometheus and Grafana for monitoring and Trivy for security scanning. A MAPE-K based adaptive engine enables continuous monitoring, analysis, planning, and execution of recovery actions, supporting self-adaptation and self-healing during performance degradation, processing failures, and data integrity issues. The project aims to demonstrate a scalable, secure, reliable, and intelligent platform for modern architectural visualization and design.

Signature of the Guide

Signature of the Course Coordinator

HOD

Tools and Technologies Used for the Project:

Category	Tools / Technologies
Frontend	React.js, Three.js, Tailwind CSS, Zustand
Backend	Python, FastAPI
Computer Vision	OpenCV, NumPy
Database	PostgreSQL
3D Visualization	Three.js
Version Control	Git, GitHub
Containerization	Docker, Docker Compose
CI/CD	GitHub Actions
Monitoring	Prometheus, Grafana
Security Scanning	Trivy
Architecture / Adaptation	MAPE-K Adaptive Engine
Testing	Pytest, Jest
API Testing	Postman
Design & Analysis	Vastu Rule Engine, Version Comparison
Development IDE	Visual Studio Code

Signature of the Guide**Course Coordinator****HOD**